

INF266 - Final Project

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Part I - MAB

Problem 1

The problem can still be formulated as a Multi-Armed Bandit (MAB), but it no longer satisfies the stationarity assumption. In the classical setting, each arm has a fixed reward distribution, whereas here the expected effectiveness changes over time due to adaptation in the population. This results in a non-stationary MAB.

An alternative is an MDP formulation where the hidden state represents population adaptation. However, since this state is not observable, the problem becomes a POMDP, which is significantly more complex. Thus, a non-stationary MAB formulation is more practical.

The main challenge is that past observations quickly become outdated. This makes it harder to estimate arm values and complicates the exploration-exploitation trade-off, as the agent must keep exploring and adapt to changes rather than converging to a fixed solution.

Problem 2

To handle time-varying rewards, the algorithm must adapt and reduce the impact of outdated data. We modify standard methods into non-stationary variants using either a sliding window or a discount factor $\gamma \in (0, 1)$ to prioritize recent observations.

We use an ϵ -greedy strategy with a constant learning rate, ensuring continuous exploration and fast adaptation. Optionally, simple change-detection mechanisms can reset estimates when performance drops.

The method assumes that recent rewards are more informative than older ones. Performance depends on parameters such as the window size or γ : small values adapt quickly but increase variance, while larger values provide stability but slower response. The key trade-off is between responsiveness and stability.

Problem 3

We compared a standard ϵ -greedy algorithm with a non-stationary variant on the `Bandits_final` environment over 1000 steps.

As shown in Figure 1, the standard method initially improves but performs worse after changes, since it averages all past rewards and adapts slowly. The non-stationary variant achieves higher average reward and recovers faster after changes by prioritizing recent observations.

Overall, adaptive methods outperform standard approaches in non-stationary environments by continuously tracking changes instead of converging prematurely.

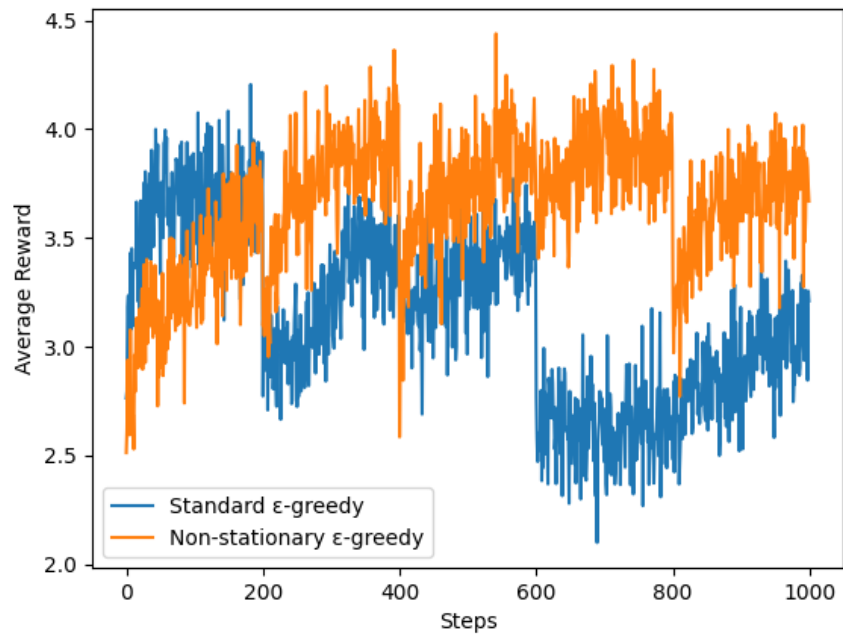


Figure 1: Comparison of standard ϵ -greedy and non-stationary ϵ -greedy over 1000 steps.